

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. ATKT Examination (CL)

April 2013

CL211 Environmental Engineering-I

Date: 26.04.2013, Friday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q-1 (a) Briefly state various factors affecting water demand. [05]
(b) Define following term (any one). [02]
(1) Hardness, (2) Most Probable Number, (3) Turbidity

- Q-2 (a) Enlist various methods used for population forecast and explain incremental increase method in detail. [05]
(b) The following is the population data of a city. Estimate the population of city in 2021 by arithmetic increase method. [05]

Year	1951	1961	1971	1981	1991	2001	2011
Population (P)	38540	52680	72100	69800	75200	98240	144700

- (c) Write short note (any one): [04]
(1) Sedimentation tank (2) Hydraulic Retention Time (3) Flocculation

OR

- Q-2 (a) For a horizontal flow rectangular sedimentation basin, prove that settling velocity of a particle is independent of basin depth. Also write the assumptions made. [05]
(b) Derive Stoke's law equation for settling velocity of particles. Also state limitations of Stoke's law. [05]
(c) Find out settling velocity of particles having specific gravity of 2.25 and diameter 0.1 mm. Take kinematic viscosity of water, ν (nu) as $1.004 \times 10^{-6} \text{ m}^2/\text{s}$. [04]

- Q-3 (a) Why coagulation is used in water treatment? Explain various mechanisms of chemical coagulation [06]
(b) With neat sketch, discuss Fill & Draw type sedimentation tank [04]
(c) A primary settling tank is 20 m in diameter with side water depth of 2.5 m. Single effluent weirs are located at the periphery of the tank. If flow of water is $20000 \text{ m}^3/\text{d}$, calculate: 1) surface area and volume, 2) surface overflow rate, 3) detention period. [04]

OR

- Q-3 (a) What is coagulation? Name a few common coagulants used and discuss one of them in detail. [06]
(b) A settling tank is designed for an overflow rate of $4 \text{ m}^3 \text{ per m}^2 \text{ per hour}$. What percentage of particles of diameter (a) 0.05 mm, and (b) 0.02 mm, will be removed in this tank at 20°C ? Assume specific gravity of particles as 2.65. [05]
(c) Differentiate between discrete settling and flocculent settling. [03]

SECTION - II

- Q-4 (a) Differentiate between a rapid sand filter and a slow sand filter. [05]
 (b) What is ventilation? [02]

- Q-5 (a) Briefly describe mechanisms of removal of suspended particles in a filter bed. [06]
 (b) Discuss various types of hardness normally found in water. [03]
 (c) Design a bell mouth canal intake for a city of 80000 persons drawing water from a canal which runs for 10 hours in a day with a depth of 1.8 m. Assume average water consumption = 140 LPCD. Assume velocity through screens and bell mouth to be less than 0.16 m/s and 0.32 m/s, respectively. [05]

OR

- Q-5 (a) What is an intake structure? Enlist various types of intake structures and describe reservoir intake with neat sketch. [06]
 (b) A filter unit is 4.0 m x 9.0 m in size. After filtering 10000 m³/d of water, the filter is backwashed at a rate of 10 L/m²/sec for 15 min. Calculate (1) average filtration rate, and (2) quantity and percentage of water used in washing. Assume that filter is stopped for 30 min. everyday for backwashing. [04]
 (c) Write short note (any one): (1) Objectives of water treatment, (2) backwashing in a rapid sand filter, (3) attributes of a good water distribution system [04]

- Q-6 (a) Name various disinfecting agents and explain various types of chlorination briefly. [06]
 (b) Write short notes (any two): (1) Air conditioning, (2) Plumbing systems in buildings, (3) Indoor air pollution, (4) Gravity and combined water distribution methods, (5) Radial water distribution system [08]

OR

- Q-6 (a) Briefly discuss various principles of an efficient drainage system [06]
 (b) Chlorine usage in a treatment plant treating 20000 m³/d of water is 7.5 kg/d. The residual chlorine after 10 min. contact is 0.2 mg/L. Calculate the chlorine dose in mg/L and chlorine demand. [04]
 (c) Write short note (any one): (1) Natural ventilation system, (2) House service connection, (3) Gully trap [04]

List of Useful formulae

$P_n = P + nI$	$P = \mu G^2 V$	$SOR = \frac{\text{Flow}}{\text{surface area}}$	$v_s = \frac{gd^2}{18\nu} (Ss - 1)$
$DT = \frac{V}{Q} = \frac{LBH}{Q}$	Pipe flow, $Q = \frac{\pi}{4} d^2 * v$	$v_s = 418(Ss - 1)d^2 \left(\frac{3t + 70}{100} \right)$	
